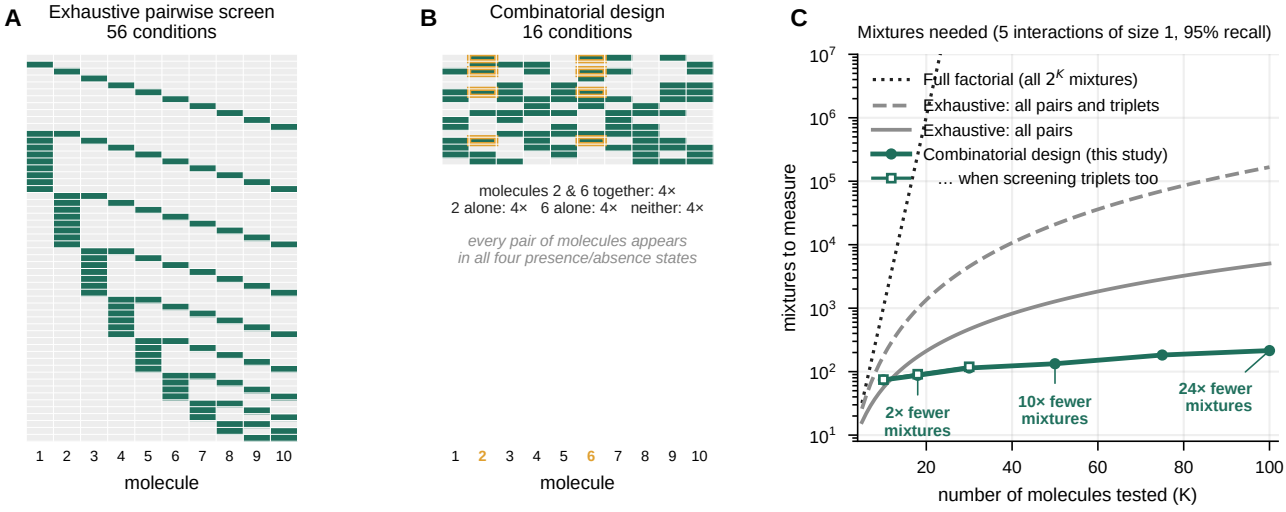
A
Exhaustive pairwise screen
56 conditions
1 2 3 4 5 6 7 8 9 10
molecule
B
Combinatorial design
16 conditions
molecules 2 & 6 together: 4×
2 alone: 4× 6 alone: 4× neither: 4×
every pair of molecules appears
in all four presence/absence states
1 2 3 4 5 6 7 8 9 10
molecule
C
Mixtures needed (5 interactions of size 1, 95% recall)
Full factorial (all 2^K mixtures)
Exhaustive: all pairs and triplets
Exhaustive: all pairs
Combinatorial design (this study)
... when screening triplets too
mixtures to measure
number of molecules tested (K)
2× fewer mixtures
10× fewer mixtures
24× fewer mixtures